

ASR Performance Report (VPB Small Dataset)

VPBank AI Center (AIC)

September 11, 2025

Introduction and Context

- **VPBank Overview:** A leading bank in Vietnam, pioneering AI-driven financial solutions for digital transformation.
- **AI Center (AIC):** Dedicated to developing cutting-edge AI technologies for banking operations.
- **Debt Collection Division (DCD):** Manages 250k customers with 25k daily calls, facing challenges in speech-to-text (STT) due to diverse user voices.
- **Collaboration with AWS:** Utilized \$14,000 in GPU credits for training and fine-tuning ASR models.
- **VPB Small Dataset:** 10 hours of sampled call data, split into 5 segments for processing and analysis.
- **Goal:** Build robust, scalable STT models to enhance automated debt collection and customer interaction.

Technical Context

- **Chunkformer (closed-source, current in use):**
 - WER 23-27%, performs well but **cannot be fine-tuned**.
- **Fast-Conformer (open-source, pre-trained on VietSpeech):**
 - WER 35-39%, underperforms.
- **Approach:** Two-step fine-tuning:
 - v1: Pseudo-labeling.
 - v2: Pseudo-labels → VPB training data.
- **Utilization of AWS Credits:** \$14,000 in GPU credits used for training and fine-tuning the Fast-Conformer model on AWS infrastructure.

WER Results (VPB Small Dataset)

Dataset	Chunkformer	v0 (VietSpeech)	v1 (Pseudo)	v2 (Pseudo→VPB)	Notes
standard_test_2	0.2499	0.3546	0.3040	0.2420	70% overlap
standard_test	0.1613	0.3378	0.2582	0.4294	Small (29 samples)
next_day_test_debug	0.2076	0.3414	0.2687	0.2649	Independent test
vpb_right2_train	0.2420	0.3633	0.2956	0.2456	Training set
vpb_right2_valid	0.2696	0.3902	0.3240	0.2821	Validation set

- Note: Metrics represent performance across 5 segments of VPB Small Dataset (10 hours).

- **Pre-trained Model:** Significantly worse than Chunkformer (WER 0.35–0.39).
- **Fine-tuning v1:** Strong improvement (WER reduction of 0.07–0.10).
- **Fine-tuning v2:** Approaches Chunkformer on **key sets (validation, next_day)**.
- **Key Notes:**
 - Low WER on **standard_test_2** due to **data leakage** (70% overlap with training).
 - Low WER on **train** indicates convergence, but **not representative of new user performance**.
- **Current Achievement:** Developed a fine-tuned Fast-Conformer model using AWS GPU resources.

Key Insights

- **Core Challenge:** Diverse user voices & noise:
 - Regional accents, speech speed, stuttering, local dialects.
 - Environmental noise, poor call signals.
- **Scale of Operations:** 250k customers in debt collection; 25k new calls daily → Major STT issues due to voice diversity vs. public datasets; even basic audio normalization (mean/std-based) fails.
- **Data Accumulation:** Plan to collect larger datasets (6,000 calls/week, 100 hours) for future labeling and processing (pseudo-labeling, self-supervised, user-based STT, etc.).
- **Model Potential:** Build robust models for unseen data (user voices).
- **Solution:** Manual labeling → Internal gold-standard dataset; continuously update with new user voices.
- **Future:** Collect user voices at loan contract stage → User verification, automated debt collection/reminders via voice-calls.

Value of Data Labeling

- **Essential for Fine-Tuning:** Required to improve and sustain performance.
- **Long-Term Data Asset:**
 - Call analysis, user verification, service quality evaluation.
 - Train NLP models.
 - Develop Callbot (automated voice interactions).
- **One-Time Investment:** Reusable across multiple iterations and applications.
- **Internal Potential:** Enhances model accuracy; builds strategic in-house data for competitive edge.

Urgency and Resource Needs

- **AWS Credits:** \$15,000 (expiring 09/20/2025) → \$14,000 already utilized for GPU training.
- **Labeling Deadline:** 09/15/2025.
- **Requirements:** 100 staff, 500 overtime hours (100 million VND).
- **Computing Needs:** Additional GPU resources for ongoing training, scaling to handle larger datasets in the future.
- **Investment Alignment:** Coordinate with AWS, DCD, and AIC for further computing + data investments.

Conclusion

- **Without Labeling:** Model stalls at 70-75% accuracy.
- **With Investment in Labeling and Computing:** Expected to achieve **75-80%+** accuracy and own **strategic gold-standard data**, with:
 - VPBank DCD providing robust data support.
 - AWS contributing additional computing resources.
- **Investment Rationale:** Leverage AWS credits, DCD collaboration, and AIC expertise for a sustainable Callbot ecosystem.
- **VPBank's Vision:** As a top-tier bank in Vietnam, VPBank leads in AI-driven banking, driving digital transformation and delivering innovative financial services.
- **Next Steps:** Align with AWS, DCD, and AIC for additional resources to advance ASR and related applications.

Acknowledgments

- **To AWS:** Heartfelt thanks for the \$14,000 GPU credits, enabling cutting-edge AI model development.
- **To DCD:** Gratitude for operational insights and collaboration, aligning AI solutions with debt collection needs.
- **To AIC:** Appreciation for relentless innovation and dedication to building world-class AI capabilities.
- **Our Commitment:** VPBank will deepen partnerships with AWS and DCD, leveraging AIC's expertise to drive AI innovation, enhance customer experiences, and maintain our position as a technology leader in banking.
- **Future Promise:** Continue delivering value to stakeholders, pushing the boundaries of AI-driven financial services, and fostering sustainable growth.